Computer Vision — 21CSE454T

Unit 4: Vision & Motion

Topic 3: Shape from Texture | SLO1

Course	Computer Vision — 21CSE454T
Unit	Unit 4: Vision & Motion
Topic	Topic 3: Shape from Texture
SLO1	Able to explain the theoretical foundations of the shape-from-texture technique.
Session Title	Texture Gradients as Depth Cues — Theoretical Foundations
Duration	10 Minutes Activity
Response Method	YES / NO cards | Thumbs 👍 / 👎 | Labelled flashcards

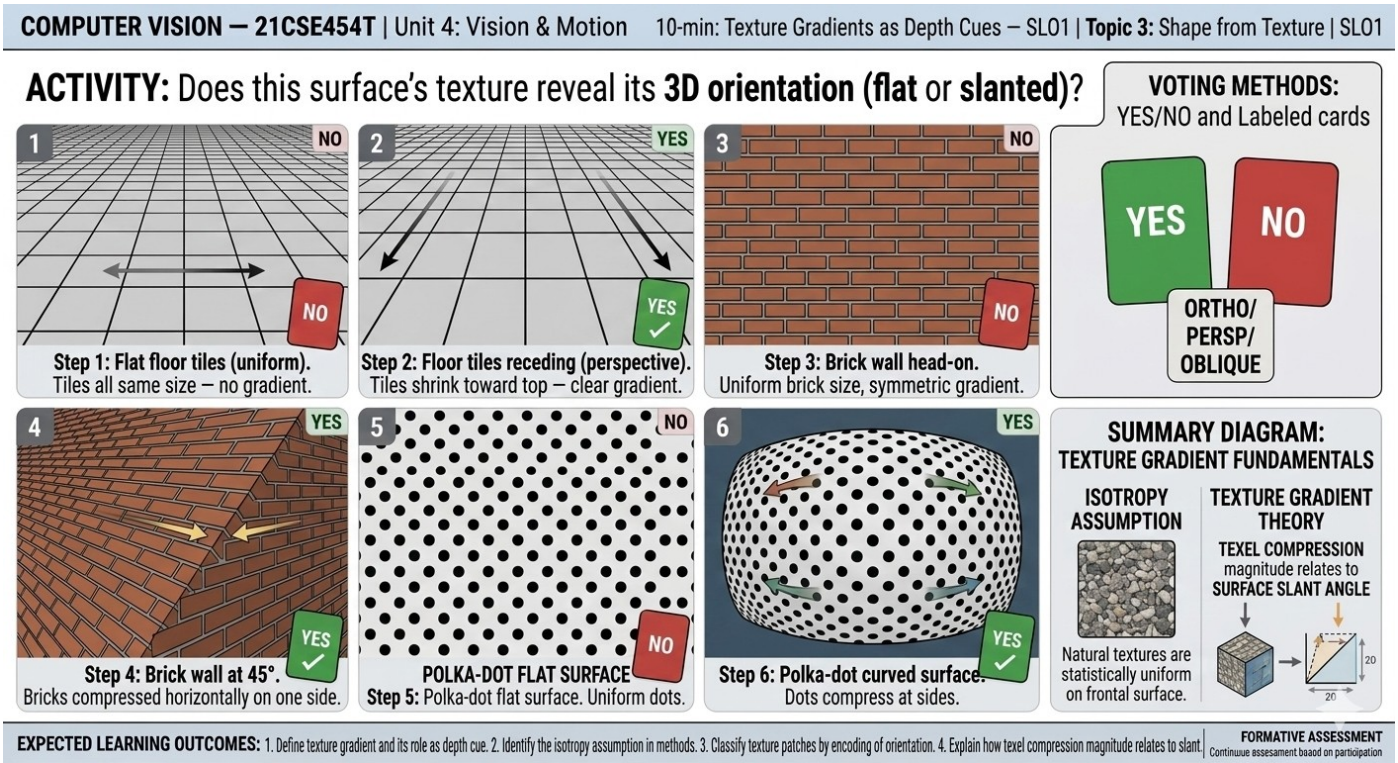
Activity Question

"Does this surface's texture reveal its 3D orientation (flat or slanted)?"

Answer: The texture gradient can reveal 3D orientation when texels change in apparent size or density. For the six surfaces: 1) NO, 2) YES, 3) NO, 4) YES, 5) NO, 6) YES.

Activity Overview

Students view 6 textured surface patches and vote on whether the texture gradient provides enough information to infer the surface's 3D orientation, reinforcing their understanding of how texture element (texel) compression encodes surface slant.



Instructor Instructions

1. Show 6 textured images: uniformly tiled floor (flat), tiled floor receding into distance (slanted), brick wall head-on, brick wall at angle, polka-dot surface flat, polka-dot surface curved.
2. Ask: 'Does the texture reveal the 3D orientation of this surface?'
3. Students vote YES / NO.
4. After each, annotate: draw arrows showing the gradient direction — where texels shrink.
5. Explain the Witkin (1981) / Stevens (1981) texture gradient theory.
6. Define isotropy assumption: natural textures are statistically uniform on a frontal surface.

Step-by-Step Activity Flow

Step	Action	Details / Expected Student Response
1	**Flat floor tiles (uniform)**	Tiles all same apparent size — no gradient. Expected: NO — surface orientation ambiguous.
2	**Floor tiles receding (perspective)**	Tiles shrink toward top of image — clear gradient. Expected: YES — slanted away.
3	**Brick wall head-on**	Uniform brick size, symmetric gradient. Expected: NO — could be flat frontal.
4	**Brick wall at 45°**	Bricks compressed horizontally on one side. Expected: YES — reveals oblique angle.
5	**Polka-dot flat surface**	Uniform dots. Expected: NO — no orientation signal.
6	**Polka-dot curved surface (cylinder)**	Dots compress at sides. Expected: YES — curvature visible from dot density change.

Student Response Method

After each image or scenario is displayed, students respond using one of:

- Raising YES card (green) or NO card (red)
- Showing thumbs up 👍 for YES or thumbs down 👎 for NO
- Holding up labelled category cards (e.g., ORTHO / PERSP / OBLIQUE)

The instructor acknowledges the class response, reveals the correct answer, and provides a 1–2 sentence explanation before moving to the next item.

Expected Learning Outcomes

#	Learning Outcome
1	Define texture gradient and explain its role as a monocular depth cue.
2	Identify the isotropy assumption underlying shape-from-texture methods.

3	Classify texture patches by whether they encode surface orientation.
4	Explain how texel compression magnitude relates to surface slant angle.